

COURSE 3 - DATA ENGINEERING 2026

C3-061 - Distributed Processing with Spark Practice Pack

Genuine-attempt tasks. Use the included starter project; review remains closed until evidence is saved.

Artifact	C3-061
Status	2026 - LEARNER EDITION
Phase	C3-05 Distributed Processing with Spark 16 lessons 46-60 focused hours
Prerequisite	C3-04 Gate PASS; C2B SQL/Python/modeling foundations assumed, distributed execution taught here.
Current lane	Apache Spark / PySpark 4.2.0 Python 3.10+ Java 17+ learner local[2]
Brand	Charcoal #1F2937 Teal #14B8A6 Soft Blue #3B82F6 AMOLED #000

House teaching rule: mental picture -> tiny worked example -> guided real Spark workflow -> expected result -> why it worked -> break/fix -> independent changed task -> validation -> physical-plan/production reasoning -> evidence. A green Spark job is not mastery if grain, quality, plan or reproducibility is wrong.

Attempt protocol

For each task: predict -> implement/inspect -> save output/plan/control -> write confidence 1-4 and one uncertainty -> only then open matching Rxx. If Spark runtime is unavailable, complete only the static/design/source-control portion and mark runtime evidence PENDING; do not fabricate screenshots or plans.

Starter ZIP

Open C3_061_PRACTICE_STARTER_PROJECT_RC1.zip. Extract it into your C3-05 module workspace. Do not edit source fixtures to make assertions pass.

P01 - Why distributed processing: partitions, executors and coordination

Draw the supplied order pipeline as driver -> partitions/tasks -> shuffle -> output. Add a second 80 MB case and decide whether Spark is justified.

Before you touch code

Write: input grain -> output grain -> expected invariant -> one likely physical consequence.

Evidence to save

- Changed code/config/design file
- Observed command/output or explicit runtime-PENDING note
- Row/key/value/rejection control
- Partition/plan evidence when relevant
- Confidence 1-4 + main uncertainty
- One sentence explaining why the result is correct, not merely successful

After saving this attempt, C3-062 R01 may be opened. Close it before the changed retry.

P02 - Spark architecture, driver and execution model

Inspect driver-safety anti-patterns and label each as safe/unsafe only after stating the maximum data that can reach the driver.

Before you touch code

Write: input grain -> output grain -> expected invariant -> one likely physical consequence.

Evidence to save

- Changed code/config/design file
- Observed command/output or explicit runtime-PENDING note
- Row/key/value/rejection control
- Partition/plan evidence when relevant
- Confidence 1-4 + main uncertainty
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After saving this attempt, C3-062 R02 may be opened. Close it before the changed retry.

P03 - PySpark DataFrames and explicit schemas

Create/verify the Spark 4.2.0 local[2] session, explicit ORDER_RAW_SCHEMA, row count and partition count. Save version and Java evidence.

Before you touch code

Write: input grain -> output grain -> expected invariant -> one likely physical consequence.

Evidence to save

- Changed code/config/design file
- Observed command/output or explicit runtime-PENDING note
- Row/key/value/rejection control
- Partition/plan evidence when relevant
- Confidence 1-4 + main uncertainty
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After saving this attempt, C3-062 R03 may be opened. Close it before the changed retry.

P04 - Selecting, filtering and expression design

Implement `clean_orders` with built-in expressions. Keep invalid rows separately and prove one clean row per valid `order_id`.

Before you touch code

Write: input grain -> output grain -> expected invariant -> one likely physical consequence.

Evidence to save

- Changed code/config/design file
- Observed command/output or explicit runtime-PENDING note
- Row/key/value/rejection control
- Partition/plan evidence when relevant
- Confidence 1-4 + main uncertainty
- One sentence explaining why the result is correct, not merely successful

After saving this attempt, C3-062 R04 may be opened. Close it before the changed retry.

P05 - Joins and distributed join consequences

Prove `products.product_id` unique, join orders -> products, and save input/output count plus join-plan evidence when runtime is available.

Before you touch code

Write: input grain -> output grain -> expected invariant -> one likely physical consequence.

Evidence to save

- Changed code/config/design file
- Observed command/output or explicit runtime-PENDING note
- Row/key/value/rejection control
- Partition/plan evidence when relevant
- Confidence 1-4 + main uncertainty
- One sentence explaining why the result is correct, not merely successful

After saving this attempt, C3-062 R05 may be opened. Close it before the changed retry.

P06 - Aggregations, grouping and window functions

Build region revenue at region grain and a customer/order sequence window at order grain. State tie-breaking rule.

Before you touch code

Write: input grain -> output grain -> expected invariant -> one likely physical consequence.

Evidence to save

- Changed code/config/design file
- Observed command/output or explicit runtime-PENDING note
- Row/key/value/rejection control
- Partition/plan evidence when relevant
- Confidence 1-4 + main uncertainty
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After saving this attempt, C3-062 R06 may be opened. Close it before the changed retry.

P07 - Nulls, types and data-cleaning patterns

Add malformed qty/timestamp cases. Produce reason-coded rejects and prove source = clean + rejected.

Before you touch code

Write: input grain -> output grain -> expected invariant -> one likely physical consequence.

Evidence to save

- Changed code/config/design file
- Observed command/output or explicit runtime-PENDING note
- Row/key/value/rejection control
- Partition/plan evidence when relevant
- Confidence 1-4 + main uncertainty
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After saving this attempt, C3-062 R07 may be opened. Close it before the changed retry.

P08 - Reading and writing CSV/JSON/Parquet/Delta

Read CSV/JSON/Parquet deliberately; write Parquet with a justified partition/file plan. Record that Delta requires its own compatible runtime/package.

Before you touch code

Write: input grain -> output grain -> expected invariant -> one likely physical consequence.

Evidence to save

- Changed code/config/design file
- Observed command/output or explicit runtime-PENDING note
- Row/key/value/rejection control
- Partition/plan evidence when relevant
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After saving this attempt, C3-062 R08 may be opened. Close it before the changed retry.

P09 - Spark SQL and DataFrame interoperability

Create a temp view and implement one metric in both Spark SQL and DataFrame API. Prove semantic parity.

Before you touch code

Write: input grain -> output grain -> expected invariant -> one likely physical consequence.

Evidence to save

- Changed code/config/design file
- Observed command/output or explicit runtime-PENDING note
- Row/key/value/rejection control
- Partition/plan evidence when relevant
- Confidence 1-4 + main uncertainty
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After saving this attempt, C3-062 R09 may be opened. Close it before the changed retry.

P10 - Partitions, repartition, coalesce and shuffles

Compare repartition and coalesce on the practice data. Save partition counts, output file counts and one Exchange/shuffle explanation.

Before you touch code

Write: input grain -> output grain -> expected invariant -> one likely physical consequence.

Evidence to save

- Changed code/config/design file
- Observed command/output or explicit runtime-PENDING note
- Row/key/value/rejection control
- Partition/plan evidence when relevant
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After saving this attempt, C3-062 R10 may be opened. Close it before the changed retry.

P11 - Query plans, explain and execution-stage reasoning

Run `explain("formatted")` on join + aggregation. Annotate Scan, Exchange, Join and Aggregate/Adaptive nodes.

Before you touch code

Write: input grain -> output grain -> expected invariant -> one likely physical consequence.

Evidence to save

- Changed code/config/design file
- Observed command/output or explicit runtime-PENDING note
- Row/key/value/rejection control
- Partition/plan evidence when relevant
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After saving this attempt, C3-062 R11 may be opened. Close it before the changed retry.

P12 - Caching and persistence decisions

Complete cache decision notes. If runtime is available, materialize one intentionally reused intermediate and unpersist it; do not cache one-pass data.

Before you touch code

Write: input grain -> output grain -> expected invariant -> one likely physical consequence.

Evidence to save

- Changed code/config/design file
- Observed command/output or explicit runtime-PENDING note
- Row/key/value/rejection control
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After saving this attempt, C3-062 R12 may be opened. Close it before the changed retry.

P13 - Data skew and uneven partition diagnosis

Profile hot keys and propose the least-invasive remedy. Repeat with an UNKNOWN-heavy changed fixture and classify quality vs valid skew.

Before you touch code

Write: input grain -> output grain -> expected invariant -> one likely physical consequence.

Evidence to save

- Changed code/config/design file
- Observed command/output or explicit runtime-PENDING note
- Row/key/value/rejection control
- Partition/plan evidence when relevant
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After saving this attempt, C3-062 R13 may be opened. Close it before the changed retry.

P14 - Adaptive execution and tuning concepts

Write one AQE hypothesis and change only one relevant setting or input distribution. Preserve correctness controls before/after.

Before you touch code

Write: input grain -> output grain -> expected invariant -> one likely physical consequence.

Evidence to save

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- Observed command/output or explicit runtime-PENDING note
- Row/key/value/rejection control
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After saving this attempt, C3-062 R14 may be opened. Close it before the changed retry.

P15 - Testing Spark transformations and output invariants

Complete PySpark tests for deterministic transform output plus schema/grain/reconciliation invariants.

Before you touch code

Write: input grain -> output grain -> expected invariant -> one likely physical consequence.

Evidence to save

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After saving this attempt, C3-062 R15 may be opened. Close it before the changed retry.

P16 - Packaging and running repeatable Spark jobs

Finish the CLI-style job, README and runbook. Prove paths/config are parameters and another clean environment can reconstruct the intended run.

Before you touch code

Write: input grain -> output grain -> expected invariant -> one likely physical consequence.

Evidence to save

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- Observed command/output or explicit runtime-PENDING note
- Row/key/value/rejection control
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After saving this attempt, C3-062 R16 may be opened. Close it before the changed retry.